

Precalculus — A Four-Week Course

Every lecture (short version) plus sample homework and exam problems with an answer key.

Week 1 — Functions and their graphs

Real numbers and intervals; functions and notation; reading graphs; transformations; combining and composing functions; inverse functions; linear functions and rates of change.

1.1 Real numbers, intervals, and absolute value

The number line that every function in this course lives on.

Real numbers, intervals, and absolute value

Precalculus works with the **real numbers**: the rationals (ratios of integers such as $3/4$ or -7) together with the **irrationals** (numbers such as $\sqrt{2}$ and π whose decimals never terminate or repeat). Every point on the number line is a real number, and every real number is a point.

Intervals and inequalities

Sets of real numbers are written as **intervals**: $[a, b]$ includes both endpoints, (a, b) excludes them, and $[a, \infty)$ runs forever to the right. Inequalities obey the rules of equations with one exception — multiplying or dividing both sides by a negative number reverses the direction of the inequality.

$$|x - a| < r \Leftrightarrow a - r < x < a + r.$$

An absolute-value inequality therefore describes an interval centered at a with radius r .

Distance

The absolute value $|a - b|$ is the distance between a and b on the line. In the plane, the Pythagorean theorem gives the **distance formula**:

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}.$$

Absolute value also obeys the **triangle inequality** $|a + b| \leq |a| + |b|$: a detour is never shorter than the direct route.

A real example

A machined piston specified as 3.500 ± 0.002 inches must satisfy $|d - 3.500| \leq 0.002$, the interval $[3.498, 3.502]$. The irrationals are just as concrete: the diagonal of a unit square has length $\sqrt{2}$, and Greek mathematicians proved that no ratio of whole numbers equals it — a discovery that forced mathematics to accept numbers beyond fractions.

1.2 Functions and notation

A rule that assigns exactly one output to each input.

Functions and notation

A **function** f from a set D to a set R assigns to each input x in D exactly one output $f(x)$. D is the **domain**, and the set of outputs actually produced is the **range**. Functions are given by formulas, tables, graphs, or words; the formula is only one representation.

Notation and evaluation

The expression $f(x)$ means the output when the input is x ; $f(a + h)$ means substitute $a + h$ everywhere x appears. The **difference quotient** measures the average rate of change of f over an interval of width h :

$$[f(x + h) - f(x)] / h.$$

It is the single most important expression in precalculus, because calculus is what happens to it as h shrinks toward zero.

Finding the domain

Unless stated otherwise, the domain is every real input for which the formula makes sense. Exclude inputs that would:

- divide by zero — $x \neq 3$ in $1/(x - 3)$;
- take an even root of a negative number — $x \geq -2$ in $\sqrt{x + 2}$;
- take the logarithm of a non-positive number — $x > 0$ in $\ln x$.

A **piecewise** function uses different formulas on different parts of its domain.

A real example

The U.S. federal income tax is a piecewise-linear function of taxable income. Each bracket applies its rate only to the dollars inside that bracket, so tax owed rises continuously: crossing into a higher bracket raises the tax on the next dollar, never on the dollars already earned. The common fear that a raise can lower take-home pay confuses the rate on the last dollar with the rate on all of them.

1.3 Graphs of functions

Reading a function's behavior from its picture.

Graphs of functions

The **graph** of f is the set of points $(x, f(x))$. A curve in the plane is the graph of a function exactly when it passes the **vertical line test**: no vertical line meets it more than once, because each input has only one output.

What a graph reveals

From a graph you can read the domain (its horizontal extent), the range (its vertical extent), the **zeros** (where it meets the x -axis), and where f is **increasing** or **decreasing**. A **local maximum** is a point higher than all nearby points; a local minimum is lower.

f increasing on $I \Rightarrow x_1 < x_2$ implies $f(x_1) < f(x_2)$.

Symmetry

Two symmetries save half the work of graphing:

- **Even** functions satisfy $f(-x) = f(x)$ — symmetric about the y-axis, like x^2 and $\cos x$.
- **Odd** functions satisfy $f(-x) = -f(x)$ — symmetric about the origin, like x^3 and $\sin x$.

Most functions are neither, but testing takes one line of algebra.

A real example

The Keeling curve plots atmospheric carbon dioxide measured at Mauna Loa since 1958. Its graph shows two behaviors at once: a long-run increase, from about 315 parts per million in 1958 to over 420 today, and a yearly oscillation with a local maximum each spring and a minimum each fall as Northern Hemisphere vegetation grows and decays. Reading a graph means separating trend from cycle.

1.4 Transformations of graphs

Shifting, stretching, and reflecting a known graph.

Transformations of graphs

Most functions in this course are built from a small library of basic graphs — x^2 , x^3 , $\sqrt{x}$, $|x|$, $1/x$, b^x , $\sin x$ — by **transformations**. Recognizing the transformation lets you sketch a graph without plotting points.

The general form

$$y = a f(b(x - h)) + k.$$

- k shifts the graph up ($k > 0$) or down;
- h shifts it right ($h > 0$) or left — note the minus sign;
- a stretches vertically by $|a|$, and $a < 0$ reflects across the x-axis;
- b compresses horizontally by $|b|$, and $b < 0$ reflects across the y-axis.

Order matters

Changes inside the function act on inputs and behave backwards; changes outside act on outputs and behave as expected. Apply horizontal changes first, then vertical stretches and reflections, then vertical shifts. The vertex of $y = 2(x - 3)^2 + 1$ is $(3, 1)$, not $(-3, 1)$.

A real example

Converting Celsius to Fahrenheit, $F = 1.8C + 32$, is a transformation of the identity function: a vertical stretch by 1.8 followed by a shift up 32. That is why the scales agree at exactly one temperature: solving $C = 1.8C + 32$ gives -40 , the one point where the transformed line crosses the original.

1.5 Combining and composing functions

Building new functions from old ones.

Combining and composing functions

Functions can be added, subtracted, multiplied, and divided pointwise: $(f + g)(x) = f(x) + g(x)$, and so on. The domain of the result is the overlap of the two domains, minus any inputs where a divisor is zero.

Composition

The most powerful combination feeds the output of one function into another:

$$(f \circ g)(x) = f(g(x)).$$

The domain of $f \circ g$ is every x in the domain of g for which $g(x)$ lies in the domain of f . Composition is generally **not commutative**: $f(g(x))$ and $g(f(x))$ are usually different functions.

Decomposition

Recognizing a complicated function as a composition is a core calculus skill. The function $\sqrt{x^2 + 1}$ is $f(g(x))$ with $g(x) = x^2 + 1$ (the inner function) and $f(u) = \sqrt{u}$ (the outer function). The chain rule of calculus differentiates exactly this structure.

A real example

A store offers 20 percent off and a \$10 coupon on a \$100 jacket. Let $g(x) = x - 10$ and $f(x) = 0.8x$. Applying the coupon first gives $f(g(100)) = \$72$; applying the discount first gives $g(f(100)) = \$70$. The order of composition is worth two dollars, and retailers write their fine print accordingly.

1.6 Inverse functions

Running a function backwards.

Inverse functions

The **inverse** of f , written f^{-1} , undoes f : if f sends a to b , then f^{-1} sends b back to a . An inverse exists only when f is **one-to-one** — distinct inputs give distinct outputs — which on a graph is the **horizontal line test**.

The defining conditions

$$f(f^{-1}(x)) = x \text{ and } f^{-1}(f(x)) = x.$$

The domain of f^{-1} is the range of f , and vice versa. To find a formula, write $y = f(x)$, swap x and y , and solve for y .

The graph of an inverse

Because inverting swaps inputs and outputs, the graph of f^{-1} is the reflection of the graph of f across the line $y = x$. A function that fails the horizontal line test can be made invertible by restricting its domain: x^2 on $[0, \infty)$ has inverse $\sqrt{x}$. This is exactly how the inverse trigonometric functions of Week 4 are defined.

A real example

The Roman historian Suetonius reports that Julius Caesar encrypted messages by shifting each letter three places down the alphabet. Encryption is a one-to-one function on letters, $E(x) = x + 3 \pmod{26}$, and decryption is its inverse, $D(x) = x - 3 \pmod{26}$. Every cipher ever used is a function that must be invertible by its intended reader and hard to invert for everyone else.

1.7 Linear functions and rates of change

Constant rate of change, and the lines that express it.

Linear functions and rates of change

A **linear function** changes by the same amount for every unit change in its input. Its graph is a line, and its constant rate of change is the **slope**.

Slope and the equations of a line

$$m = (y_2 - y_1) / (x_2 - x_1).$$

A line can be written in **slope-intercept form** $y = mx + b$ or in **point-slope form**:

$$y - y_1 = m(x - x_1).$$

Horizontal lines have slope 0; vertical lines have undefined slope and are not functions.

Parallel and perpendicular lines

Parallel lines have equal slopes. Perpendicular lines have slopes whose product is -1 :

$$m_1 m_2 = -1.$$

For a nonlinear function, the slope of the line through two points of its graph is its **average rate of change** — the difference quotient of Lecture 1.2.

A real example

Road grade is slope expressed as a percentage. Baldwin Street in Dunedin, New Zealand, long certified by Guinness as the world's steepest residential street, has a maximum gradient of about 35 percent — it rises roughly 35 feet for every 100 feet of horizontal distance, a slope of 0.35. By comparison, U.S. interstate highways rarely exceed 6 percent.

Week 2 — Polynomial and rational functions

Quadratic functions; polynomial functions and their graphs; polynomial division; zeros of polynomials; complex numbers; rational functions and asymptotes; polynomial and rational inequalities.

2.1 Quadratic functions

The parabola, its vertex, and the quadratic formula.

Quadratic functions

A **quadratic function** has the form $f(x) = ax^2 + bx + c$ with $a \neq 0$. Its graph is a **parabola** opening up if $a > 0$ and down if $a < 0$, with a single turning point, the **vertex**.

Vertex form

Completing the square rewrites any quadratic as

$$f(x) = a(x - h)^2 + k,$$

with vertex (h, k) and $h = -b / (2a)$. The vertex gives the maximum or minimum value of f , which makes quadratics the first optimization tool of the course.

The quadratic formula

The zeros of $ax^2 + bx + c$ are

$$x = (-b \pm \sqrt{b^2 - 4ac}) / (2a).$$

The **discriminant** $D = b^2 - 4ac$ decides their nature:

- $D > 0$ — two real zeros;
- $D = 0$ — one repeated real zero;
- $D < 0$ — two complex conjugate zeros.

A real example

Galileo showed around 1638 that, ignoring air resistance, a projectile follows a parabola. A ball thrown straight up at 64 feet per second has height $h(t) = -16t^2 + 64t$. The vertex is at $t = -64/(2 \cdot (-16)) = 2$ seconds, where $h = 64$ feet, and the ball lands when $h = 0$, at $t = 4$ — symmetric about the vertex, as every parabola is.

2.2 Polynomial functions and their graphs

Smooth curves determined by their degree and their zeros.

Polynomial functions and their graphs

A **polynomial** of degree n is $f(x) = a_n x^n + \dots + a_1 x + a_0$ with $a_n \neq 0$. Polynomial graphs are smooth and unbroken — no gaps, jumps, or corners.

End behavior and turning points

For large $|x|$ the leading term dominates, so end behavior depends only on the degree and the sign of a_n :

$$f(x) \approx a_n x^n \text{ as } |x| \rightarrow \infty.$$

Even degree: both ends point the same way. Odd degree: the ends point opposite ways. A degree- n polynomial has at most n real zeros and at most $n - 1$ turning points.

Zeros and multiplicity

If $(x - c)^k$ is a factor, c is a zero of **multiplicity** k . At a zero of odd multiplicity the graph crosses the x -axis; at even multiplicity it touches and turns back. The factored form $f(x) = (x + 2)(x - 1)^2$ thus sketches itself: cross at -2 , bounce at 1 , rise to the right.

A real example

Every letter on this page is drawn with polynomials. Digital fonts store glyph outlines as chains of low-degree polynomial curves — PostScript fonts use cubic Bézier curves, TrueType fonts use quadratic ones — and the printer or screen evaluates them at whatever size is needed. Polynomials are cheap to compute and smooth to look at, which is why they draw nearly all of modern typography.

2.3 Polynomial division

Dividing polynomials, and the remainder and factor theorems.

Polynomial division

Polynomials can be divided just as integers can. Long division — or its shortcut, **synthetic division**, when the divisor is $x - c$ — produces a quotient and a remainder of lower degree than the divisor.

The division algorithm

$$f(x) = d(x) q(x) + r(x), \deg r < \deg d.$$

Dividing by $x - c$ leaves a constant remainder, and substituting $x = c$ gives the **remainder theorem**:

$$f(x) = (x - c) q(x) + f(c).$$

The factor theorem

It follows at once that $x - c$ is a factor of f exactly when $f(c) = 0$. Zeros and linear factors are the same information. Once one zero is found, dividing it out leaves a polynomial of lower degree — the standard strategy for factoring a cubic or quartic one root at a time.

A real example

Synthetic division is also the fastest way to evaluate a polynomial. Evaluating a degree- n polynomial this way — Horner's method, published by William Horner in 1819 and known to the Chinese mathematician Qin Jiushao in the thirteenth century — takes only n multiplications and n additions. It is still the method computers use to evaluate polynomials today.

2.4 Zeros of polynomials

Where polynomial equations have their solutions, and how to find them.

Zeros of polynomials

Finding the zeros of a polynomial is solving the equation $f(x) = 0$. For degree two the quadratic formula suffices; beyond that, a set of theorems narrows the search.

The rational zeros theorem

If f has integer coefficients, every rational zero p/q in lowest terms has p dividing the constant term and q dividing the leading coefficient:

$$p \mid a_0, q \mid a_n.$$

This produces a finite list of candidates to test with synthetic division.

The fundamental theorem of algebra

Every polynomial of degree $n \geq 1$ has exactly n complex zeros, counted with multiplicity:

$$f(x) = a_n(x - c_1)(x - c_2) \dots (x - c_n).$$

Carl Friedrich Gauss gave the first substantial proof in his 1799 doctoral dissertation. For real coefficients, nonreal zeros come in **conjugate pairs** $a \pm bi$.

A real example

Formulas for the cubic and quartic were published by Gerolamo Cardano in 1545. For three centuries mathematicians sought a formula for the quintic; in 1824 Niels Henrik Abel proved none exists in radicals. The fundamental theorem guarantees five zeros exist — Abel showed they generally cannot be written with roots, which is why numerical methods, not formulas, find them in practice.

2.5 Complex numbers

Extending the number line so every polynomial has its zeros.

Complex numbers

The equation $x^2 = -1$ has no real solution, so mathematics adjoins one: the **imaginary unit** i , whose square is -1 .

Arithmetic

$$i^2 = -1.$$

A **complex number** is $a + bi$ with a, b real. Add and multiply as with polynomials, replacing i^2 with -1 . The **conjugate** of $a + bi$ is $a - bi$, and a number times its conjugate is real: $(a + bi)(a - bi) = a^2 + b^2$. To divide, multiply top and bottom by the conjugate of the denominator.

The complex plane

Plot $a + bi$ as the point (a, b) . Its distance from the origin is the **modulus**:

$$|a + bi| = \sqrt{a^2 + b^2}.$$

Addition is vector addition; multiplication, as Week 4 shows, multiplies lengths and adds angles.

A real example

Electrical engineers use complex numbers daily. In an alternating-current circuit, a resistor, capacitor, or inductor is described by a complex **impedance** $Z = R + jX$ — engineers write j instead of i because i already means current. Ohm's law $V = IZ$ then handles phase shifts automatically, turning differential equations into complex arithmetic.

2.6 Rational functions and asymptotes

Quotients of polynomials, and the lines their graphs approach.

Rational functions and asymptotes

A **rational function** is a quotient $f(x) = p(x)/q(x)$ of polynomials. Its domain excludes the zeros of q , and near those points the graph can blow up.

Vertical asymptotes and holes

After canceling common factors, each remaining zero of the denominator gives a **vertical asymptote** $x = c$, where $|f(x)| \rightarrow \infty$. A factor that cancels leaves a **hole** instead: $(x^2 - 1)/(x - 1)$ equals $x + 1$ everywhere except at $x = 1$.

Horizontal asymptotes

Compare degrees n (numerator) and m (denominator):

- $n < m$ — horizontal asymptote $y = 0$;
- $n = m$ — horizontal asymptote $y = a_n/b_m$, the ratio of leading coefficients;
- $n = m + 1$ — a slant asymptote, found by division.

$$y = a_n / b_m \text{ when } n = m.$$

A real example

The thin-lens equation of optics, $1/f = 1/d_o + 1/d_i$, gives image distance $d_i = f d_o / (d_o - f)$, a rational function of object distance. It has a vertical asymptote at $d_o = f$: an object at the focal point produces no image at all, because the rays emerge parallel. That is precisely how a lighthouse or flashlight turns a bulb into a beam.

2.7 Polynomial and rational inequalities

Solving inequalities by tracking where the sign changes.

Polynomial and rational inequalities

To solve $p(x) > 0$ or $p(x)/q(x) \leq 0$, move everything to one side and ask where the expression is positive or negative. Polynomials and rational functions can change sign only at their zeros and at the zeros of their denominators.

The sign chart

Find the **critical numbers** — zeros of the numerator and denominator — and mark them on the number line. On each interval between them the sign is constant, so one test point per interval settles it:

$$(x - 1)(x - 3) < 0 \Rightarrow 1 < x < 3.$$

Common errors

- Never multiply both sides by an expression whose sign is unknown — it may reverse the inequality.
- Zeros of the denominator are never included, even with $\leq$ or $\geq$.
- At a zero of even multiplicity the sign does not change.

Graphing the function and reading off where it lies above or below the axis is a useful check.

A real example

The ball of Lecture 2.1, with $h(t) = -16t^2 + 64t$, is above 48 feet when $-16t^2 + 64t > 48$, which simplifies to $t^2 - 4t + 3 < 0$, or $(t - 1)(t - 3) < 0$. The sign chart gives $1 < t < 3$: the ball spends exactly two seconds above 48 feet, one second rising and one falling.

Week 3 — Exponential, logarithmic, and trigonometric functions

Exponential functions; logarithmic functions; properties of logarithms; growth and decay models; angles and radian measure; the unit circle; graphs of sine and cosine.

3.1 Exponential functions

Growth by a constant factor rather than a constant amount.

Exponential functions

A linear function adds the same amount per unit of input; an **exponential function** $f(x) = b^x$ ($b > 0$, $b \neq 1$) multiplies by the same factor. For $b > 1$ it grows; for $0 < b < 1$ it decays. Its domain is all reals, its range is $(0, \infty)$, and its graph has the horizontal asymptote $y = 0$.

Compound interest

A principal P at annual rate r compounded n times a year for t years grows to

$$A = P(1 + r/n)^{nt}.$$

As n increases without bound, compounding becomes continuous.

The number e

The limit of $(1 + 1/n)^n$ as $n \rightarrow \infty$ is the constant $e \approx 2.71828$, and continuous compounding gives

$$A = Pe^{rt}.$$

The natural exponential e^x is the base calculus prefers, because it is the one function equal to its own rate of change.

A real example

Jacob Bernoulli discovered e in 1683 while studying compound interest. One dollar at 100 percent annual interest becomes \$2 compounded yearly, about \$2.61 monthly, about \$2.71 daily — and never more than $e \approx \$2.718$, however often interest is compounded. More frequent compounding helps, but with sharply diminishing returns.

3.2 Logarithmic functions

The inverse of exponentiation.

Logarithmic functions

Because b^x is one-to-one, it has an inverse, the **logarithm** base b . The logarithm answers the question: to what power must b be raised to give x ?

Definition

$$y = \log_b x \Leftrightarrow b^y = x.$$

The **common logarithm** $\log x$ uses base 10; the **natural logarithm** $\ln x$ uses base e . The domain of $\log_b x$ is $(0, \infty)$, its range is all reals, and its graph — the reflection of b^x across $y = x$ — has a vertical asymptote at $x = 0$.

Logarithmic scales

Logarithms compress quantities spanning many orders of magnitude into manageable numbers. Each unit on a log scale represents a constant factor, not a constant amount:

- decibels — 10 dB is a tenfold increase in sound intensity;
- pH — each unit is a tenfold change in hydrogen-ion concentration;
- earthquake magnitude — each unit is a tenfold increase in wave amplitude.

A real example

The 2011 Tōhoku earthquake off Japan registered magnitude about 9.1. Each whole step in magnitude multiplies measured amplitude by 10 and released energy by roughly 32, so a magnitude 9 quake releases about 1,000 times the energy of a magnitude 7. The logarithmic scale turns a range of trillions into a range of single digits.

3.3 Properties of logarithms

Turning multiplication into addition, and solving exponential equations.

Properties of logarithms

Because logarithms are exponents, the laws of exponents become the laws of logarithms. These laws are what made logarithms the calculating tool of science for three centuries.

The laws

- Product: $\log_b(MN) = \log_b M + \log_b N$.
- Quotient: $\log_b(M/N) = \log_b M - \log_b N$.
- Power: $\log_b(M^p) = p \log_b M$.

Any logarithm can be computed from natural logs by the **change-of-base formula**:

$$\log_b x = \ln x / \ln b.$$

Solving exponential equations

To solve an equation with the unknown in an exponent, isolate the exponential and take logarithms of both sides. For $5 \cdot 2^x = 40$, divide to get $2^x = 8$ and conclude $x = 3$; for $3^x = 20$, $x = \ln 20 / \ln 3 \approx 2.73$. Always check that solutions of logarithmic equations lie in the domain.

A real example

John Napier published logarithms in 1614, and within years astronomers — Kepler among them — used tables to replace laborious multiplications with additions. Pierre-Simon Laplace later remarked that logarithms doubled the working life of astronomers. The slide rule, two logarithmic scales sliding past each other, put the product law in every engineer's pocket until the 1970s.

3.4 Exponential growth and decay

Modeling quantities that change in proportion to their size.

Exponential growth and decay

When a quantity changes at a rate proportional to its current size — a population, an investment, a radioactive sample — it follows an exponential model.

The model

$$N(t) = N_0 e^{kt},$$

where N_0 is the initial amount and k the continuous rate: $k > 0$ for growth, $k < 0$ for decay. Two data points determine k by solving an exponential equation.

Doubling time and half-life

Exponential quantities double (or halve) over fixed intervals regardless of their size. For decay with rate $k < 0$, the **half-life** is

$$t_{1/2} = \ln 2 / |k|,$$

and the doubling time for growth is $\ln 2 / k \approx 0.693/k$ — the source of the familiar rule of 70 for growth rates quoted in percent.

A real example

Carbon-14 decays with a half-life of about 5,730 years. Living organisms maintain a steady ratio of carbon-14 to carbon-12; at death the carbon-14 begins to decay. Radiocarbon dating of Ötzi the Iceman, found in the Alps in 1991, showed he died roughly 5,300 years ago — the remaining carbon-14 fixed the age by solving $N(t) = N_0 e^{kt}$ for t .

3.5 Angles and radian measure

Measuring angles by the arcs they cut off.

Angles and radian measure

An angle is a rotation from an initial ray to a terminal ray; counterclockwise rotation is positive. Degrees divide a full turn into 360 parts, an arbitrary Babylonian convention.

Radians measure an angle by the arc it subtends on a circle, divided by the radius.

Radian measure

$$\theta = s / r, \pi \text{ rad} = 180^\circ.$$

A full turn is 2π radians. Radians are pure numbers — a ratio of two lengths — which is why calculus requires them: only in radians does $\sin x \approx x$ for small x .

Arc length, sector area, and speed

With θ in radians:

- arc length $s = r\theta$;
- sector area $A = \frac{1}{2} r^2 \theta$;
- linear speed $v = r\omega$, where ω is angular speed in radians per unit time.

$$s = r\theta.$$

A real example

Around 240 BCE Eratosthenes noted that at noon on the summer solstice the sun stood directly overhead at Syene, while at Alexandria, about 5,000 stadia north, it was about 7.2° — one-fiftieth of a circle — from vertical. Treating the distance as an arc, $s = r\theta$ gave Earth's circumference as $50 \times 5,000 = 250,000$ stadia, remarkably close to the modern value.

3.6 The unit circle and the trigonometric functions

Sine and cosine as coordinates of a point moving around a circle.

The unit circle and the trigonometric functions

Place an angle θ in standard position and let its terminal ray meet the **unit circle** $x^2 + y^2 = 1$ at the point (x, y) . The trigonometric functions are defined by that point:

Definitions

$$\cos \theta = x, \sin \theta = y, \tan \theta = y / x.$$

The reciprocals are $\sec \theta = 1/\cos \theta$, $\csc \theta = 1/\sin \theta$, and $\cot \theta = 1/\tan \theta$. For an acute angle in a right triangle, these reduce to the familiar ratios: sine is opposite over hypotenuse, cosine adjacent over hypotenuse, tangent opposite over adjacent.

Key values and the fundamental identity

The special angles $\pi/6$, $\pi/4$, and $\pi/3$ have exact values built from $1/2$, $\sqrt{2}/2$, and $\sqrt{3}/2$. Because (x, y) lies on the unit circle,

$$\sin^2 \theta + \cos^2 \theta = 1.$$

Signs in each quadrant follow from the signs of x and y , and values repeat every 2π .

A real example

The Great Trigonometrical Survey of India measured the heights of Himalayan peaks from stations more than 100 miles away, using angles of elevation and the tangent function. In 1856 it announced that Peak XV — Mount Everest — stood 29,002 feet high. The 2020 Chinese-Nepali survey found 8,848.86 meters, about 29,032 feet: nineteenth-century trigonometry was off by roughly one-tenth of one percent.

3.7 Graphs of sine and cosine

The functions of every periodic phenomenon.

Graphs of sine and cosine

As θ runs around the unit circle, $\sin \theta$ and $\cos \theta$ oscillate between -1 and 1 , repeating every 2π . Their graphs are **sinusoids**, identical except that cosine is sine shifted left by $\pi/2$.

The general sinusoid

$$y = A \sin(B(x - C)) + D.$$

- **amplitude** $|A|$ — half the distance from peak to trough;
- **period** $2\pi/B$ — the length of one cycle;
- **phase shift** C — the horizontal shift;
- **midline** $y = D$ — the center of the oscillation.

Other trigonometric graphs

The tangent function has period π and vertical asymptotes wherever $\cos x = 0$. Its graph rises through every real value on each period, which is why $\tan x$ has an inverse defined on all reals. Secant and cosecant, as reciprocals, have asymptotes at the zeros of cosine and sine.

A real example

U.S. household electricity is nominally 120 volts at 60 hertz. The 120 is a root-mean-square average; the actual voltage is the sinusoid $V(t) \approx 170 \sin(120\pi t)$, with amplitude $120\sqrt{2} \approx 170$ volts and period $1/60$ second. Every outlet in the country delivers a sine wave that swings from $+170$ to -170 volts sixty times a second.

Week 4 — Trigonometry, analytic geometry, and the road to calculus

Trigonometric identities; inverse trigonometric functions and equations; the laws of sines and cosines; polar coordinates and complex numbers; conic sections; vectors; sequences and series; a capstone synthesis.

4.1 Trigonometric identities

Equations that hold for every angle, and how to use them.

Trigonometric identities

An **identity** is an equation true for every value of the variable. Trigonometric identities let you rewrite an expression into whatever form a problem requires — the essential skill for simplifying, solving equations, and later integrating.

The core identities

- Pythagorean: $\sin^2 \theta + \cos^2 \theta = 1$, and its forms $1 + \tan^2 \theta = \sec^2 \theta$;
- Even/odd: $\cos(-\theta) = \cos \theta$, $\sin(-\theta) = -\sin \theta$;
- Cofunction: $\sin(\pi/2 - \theta) = \cos \theta$.

The **sum formulas** are the source of nearly all the rest:

$$\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta.$$

$$\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta.$$

Double-angle and half-angle formulas

Setting $\alpha = \beta$ gives $\sin 2\theta = 2 \sin \theta \cos \theta$ and $\cos 2\theta = \cos^2 \theta - \sin^2 \theta = 1 - 2 \sin^2 \theta$.

Solving the latter for $\sin^2 \theta$ yields the power-reduction formulas used constantly in calculus. To verify an identity, transform one side only, usually the more complicated one.

A real example

Around 150 CE Ptolemy compiled in the *Almagest* a table of chords — equivalent to a sine table — at half-degree intervals. He built it from a handful of known values using a theorem about quadrilaterals inscribed in a circle that is equivalent to the sum and difference formulas. The identities were the computational engine of astronomy for over a thousand years.

4.2 Inverse trigonometric functions and trigonometric equations

Recovering angles from ratios.

Inverse trigonometric functions and trigonometric equations

Sine, cosine, and tangent are not one-to-one, so their inverses are defined on restricted domains:

The principal inverses

- $\arcsin x$ has range $[-\pi/2, \pi/2]$;
- $\arccos x$ has range $[0, \pi]$;
- $\arctan x$ has range $(-\pi/2, \pi/2)$.

$$y = \arcsin x \Leftrightarrow \sin y = x, -\pi/2 \leq y \leq \pi/2.$$

The calculator returns only the principal value; other solutions must be found by symmetry and periodicity.

Solving equations

Because sine has period 2π and $\sin(\pi - \theta) = \sin \theta$, the equation $\sin x = k$ ($|k| \leq 1$) has the general solution

$$x = \arcsin k + 2\pi n \text{ or } x = \pi - \arcsin k + 2\pi n.$$

Isolate the trigonometric function first, using identities or factoring as needed, then list all solutions in the required interval.

A real example

Snell's law of refraction, $n_1 \sin \theta_1 = n_2 \sin \theta_2$, determines how light bends between media. Light traveling from water ($n \approx 1.33$) into air cannot escape if it strikes the surface beyond the critical angle $\arcsin(1/1.33) \approx 48.8^\circ$; it reflects back entirely. Fiber-optic cables trap light by exactly this total internal reflection, carrying most of the world's internet traffic.

4.3 The laws of sines and cosines

Solving triangles that are not right triangles.

The laws of sines and cosines

Right-triangle trigonometry handles only right angles. Two laws extend it to every triangle with sides a, b, c opposite angles A, B, C .

The law of sines

$$a / \sin A = b / \sin B = c / \sin C.$$

Use it when you know two angles and a side, or two sides and an angle opposite one of them. The second case is **ambiguous**: the given data may fit zero, one, or two triangles.

The law of cosines

$$c^2 = a^2 + b^2 - 2ab \cos C.$$

Use it when you know three sides, or two sides and the included angle. When $C = 90^\circ$ it reduces to the Pythagorean theorem. The area of any triangle is $\frac{1}{2}ab \sin C$.

A real example

The meter was originally defined as one ten-millionth of the distance from the North Pole to the equator. Between 1792 and 1798 Jean-Baptiste Delambre and Pierre Méchain measured the meridian arc from Dunkirk to Barcelona through a chain of more than a hundred triangles, measuring angles at each station and solving the triangles with the law of sines. Trigonometry literally defined the unit of length.

4.4 Polar coordinates and complex numbers

Locating points by distance and direction.

Polar coordinates and complex numbers

In **polar coordinates** a point is given by its distance r from the origin and the angle θ from the positive x -axis. Some curves — circles, spirals, roses — have far simpler equations in polar form.

Converting

$$x = r \cos \theta, y = r \sin \theta, r^2 = x^2 + y^2.$$

The same point has infinitely many polar representations, since adding 2π to θ changes nothing.

Complex numbers in polar form

A complex number $z = a + bi$ can be written $z = r(\cos \theta + i \sin \theta)$, where $r = |z|$. In this form multiplication multiplies moduli and adds angles, and repeated multiplication gives

De Moivre's theorem:

$$[r(\cos \theta + i \sin \theta)]^n = r^n(\cos n\theta + i \sin n\theta).$$

Euler's formula $e^{i\theta} = \cos \theta + i \sin \theta$ packages all of this, and at $\theta = \pi$ gives $e^{i\pi} + 1 = 0$.

A real example

A radar screen is a polar graph. The antenna sweeps through angle θ , and each echo is plotted at distance r proportional to its return time — the plan position indicator used by air-traffic control and weather radar since the 1940s. Converting those polar readings to x - y map coordinates is exactly $x = r \cos \theta$, $y = r \sin \theta$.

4.5 Conic sections

The curves cut from a cone by a plane.

Conic sections

Slicing a double cone with a plane yields the **conic sections**: circles, ellipses, parabolas, and hyperbolas. Each is also the graph of a second-degree equation in x and y .

Standard forms

- Circle: $(x - h)^2 + (y - k)^2 = r^2$;
- Parabola: $y^2 = 4px$, the set of points equidistant from a focus and a directrix;

- Ellipse: points whose distances to two foci sum to a constant;
- Hyperbola: $x^2/a^2 - y^2/b^2 = 1$, points whose distances to two foci differ by a constant.

$$x^2/a^2 + y^2/b^2 = 1.$$

Eccentricity

The **eccentricity** $e = c/a$, where c is the center-to-focus distance, measures how far a conic departs from a circle: $e = 0$ for a circle, $0 < e < 1$ for an ellipse, $e = 1$ for a parabola, and $e > 1$ for a hyperbola. The reflective property of the parabola — rays parallel to its axis reflect through the focus — explains satellite dishes and headlights.

A real example

In 1609 Johannes Kepler announced that planets move in ellipses with the Sun at one focus, overturning two thousand years of circular orbits. Earth's orbit has eccentricity about 0.017, nearly a circle; Halley's comet, with eccentricity about 0.967, is a long thin ellipse that brings it close to the Sun only once every 75 or 76 years.

4.6 Vectors in the plane

Quantities with both size and direction.

Vectors in the plane

A **vector** has magnitude and direction — velocity, force, displacement — as opposed to a **scalar**, which has magnitude only. In components, $\mathbf{v} = \langle v_1, v_2 \rangle$.

Operations

Vectors add component by component, which geometrically is the tip-to-tail rule, and scalar multiplication stretches them. The magnitude is

$$|\mathbf{v}| = \sqrt{v_1^2 + v_2^2},$$

and any vector can be written as $|\mathbf{v}|(\cos \theta, \sin \theta)$, linking vectors directly to the unit circle.

The dot product

$$\mathbf{u} \cdot \mathbf{v} = u_1 v_1 + u_2 v_2 = |\mathbf{u}| |\mathbf{v}| \cos \theta.$$

The dot product measures how much two vectors point the same way. It gives the angle between vectors, tests perpendicularity ($\mathbf{u} \cdot \mathbf{v} = 0$), and computes work: $W = \mathbf{F} \cdot \mathbf{d}$.

A real example

A plane flying north at 200 knots through a 30-knot wind blowing east has ground velocity $\langle 30, 200 \rangle$. Its ground speed is $\sqrt{30^2 + 200^2} \approx 202$ knots, and it drifts $\arctan(30/200) \approx 8.5^\circ$ east of its heading. Pilots correct by pointing the nose into the wind — the wind-correction angle every flight plan computes.

4.7 Sequences and series

Ordered lists of numbers, and what happens when you add them.

Sequences and series

A **sequence** is a function whose domain is the positive integers, written $a_1, a_2, a_3, \dots$. A **series** is the sum of its terms.

Arithmetic and geometric sequences

An arithmetic sequence adds a constant difference d ; a geometric sequence multiplies by a constant ratio r :

$$a_n = a_1 + (n - 1)d, \quad a_n = a_1 r^{n-1}.$$

The finite sums are $S_n = n(a_1 + a_n)/2$ for arithmetic and $S_n = a_1(1 - r^n)/(1 - r)$ for geometric.

Infinite geometric series

When $|r| < 1$, the powers r^n shrink to zero and the infinite sum converges:

$$S = a_1 / (1 - r), \quad |r| < 1.$$

When $|r| \geq 1$ the series diverges. This is the first appearance of a limit — the idea on which all of calculus rests.

A real example

A famous story holds that the young Carl Friedrich Gauss, told to add the integers from 1 to 100, paired 1 with 100, 2 with 99, and so on: fifty pairs each summing to 101, for a total of 5,050. The arithmetic-series formula is his trick in general form. The repeating decimal $0.999\dots$ is the geometric series $9/10 + 9/100 + \dots$, whose sum is $(9/10)/(1 - 1/10) = 1$ exactly.

4.8 Capstone synthesis

One idea behind everything: functions, and how they change.

Capstone synthesis

Step back, and the entire course reduces to a single sentence: precalculus is the study of functions — how to build them, read them, invert them, and measure how they change.

The one expression

$$[f(x + h) - f(x)] / h.$$

Everything you have learned prepares you to handle this quotient for a different family of functions:

- Linear functions have a constant difference quotient — their slope (Week 1).
- Polynomials and rationals simplify it by algebra and division (Week 2).
- Exponentials and logarithms require their laws to rewrite it (Week 3).
- Trigonometric functions require the sum identities (Week 4).

Where calculus begins

The difference quotient gives the average rate of change over an interval of width h . Calculus asks what happens as $h \rightarrow 0$ — the **limit** — and that limit is the instantaneous rate of change, the derivative. The infinite geometric series of Lecture 4.7 was the first such limit; calculus makes the idea systematic.

A real example

A stone dropped from a cliff falls $s(t) = 16t^2$ feet in t seconds. Over the interval from 2 to $2 + h$ seconds its average speed is $[16(2 + h)^2 - 64]/h = 64 + 16h$ feet per second. As h shrinks toward zero this approaches 64 feet per second — the stone's speed at the instant $t = 2$. That calculation is the first step of calculus, and everything in this course was needed to make it.

Practice problems

A sample of the graded work in the course: two homeworks plus the midterm and final. Answers follow at the end.

Homework 1.1 — Functions and their graphs

Short-answer problems on functions. Write each answer as the key formula in symbols using the on-screen math keyboard (use the Algebra and Trigonometry tabs for exponents, radicals, and Greek letters).

1. Write the distance formula between (x_1, y_1) and (x_2, y_2) .
2. Write the slope of the line through (x_1, y_1) and (x_2, y_2) .
3. Write the point-slope form of the line with slope m through (x_1, y_1) .
4. Write the difference quotient of a function f .

Homework 1.2 — Transformations, composition, and inverses

Use the math keyboard for exponents, parentheses, and the composition symbol $\circ$.

1. Write the transformation of $y = f(x)$ with vertical factor a , horizontal factor b , and shifts h and k .
2. Write the composition $(f \circ g)(x)$.
3. Write the two conditions that define the inverse function f^{-1} .
4. Write the relationship between the slopes m_1 and m_2 of perpendicular lines.

Midterm — Weeks 1 & 2

Cumulative midterm on functions and on polynomial and rational functions. 60 minutes. Math keyboard available; pasting disabled.

1. Write the difference quotient of a function f .
2. Write the slope of the line through (x_1, y_1) and (x_2, y_2) .
3. Write the vertex form of a quadratic with vertex (h, k) .
4. Write the quadratic formula for $ax^2 + bx + c = 0$.
5. Write the discriminant D of $ax^2 + bx + c$.
6. Write the remainder theorem for division of $f(x)$ by $x - c$.
7. Write the defining property of the imaginary unit i .
8. Write the modulus of the complex number $a + bi$.

Final Exam — Precalculus

Cumulative final covering all four weeks. 90 minutes. Math keyboard available; pasting disabled.

1. Write the quadratic formula for $ax^2 + bx + c = 0$.
2. Write the composition $(f \circ g)(x)$.
3. Write the continuous-compounding formula with principal P , rate r , and time t .
4. Write the change-of-base formula for $\log_b x$ in terms of natural logarithms.
5. Write the half-life $t_{1/2}$ for decay at continuous rate k .
6. Write arc length s in terms of radius r and angle θ in radians.
7. Write the Pythagorean identity.
8. Write the law of cosines for side c opposite angle C .
9. Write De Moivre's theorem for $[r(\cos \theta + i \sin \theta)]^n$.
10. Write the sum S of an infinite geometric series with first term a_1 and ratio r , $|r| < 1$.

Answer key

Homework 1.1 — Functions and their graphs:

1. $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$
2. $m = (y_2 - y_1) / (x_2 - x_1)$
3. $y - y_1 = m(x - x_1)$
4. $[f(x + h) - f(x)] / h$

Homework 1.2 — Transformations, composition, and inverses:

1. $y = a f(b(x - h)) + k$
2. $(f \circ g)(x) = f(g(x))$
3. $f(f^{-1}(x)) = x$ and $f^{-1}(f(x)) = x$
4. $m_1 m_2 = -1$

Midterm — Weeks 1 & 2:

1. $[f(x + h) - f(x)] / h$
2. $m = (y_2 - y_1) / (x_2 - x_1)$
3. $f(x) = a(x - h)^2 + k$
4. $x = (-b \pm \sqrt{b^2 - 4ac}) / (2a)$
5. $D = b^2 - 4ac$
6. $f(x) = (x - c) q(x) + f(c)$
7. $i^2 = -1$
8. $|a + bi| = \sqrt{a^2 + b^2}$

Final Exam — Precalculus:

1. $x = (-b \pm \sqrt{b^2 - 4ac}) / (2a)$
2. $(f \circ g)(x) = f(g(x))$
3. $A = Pe^{(rt)}$
4. $\log_b x = \ln x / \ln b$
5. $t_{(1/2)} = \ln 2 / |k|$
6. $s = r\theta$
7. $\sin^2\theta + \cos^2\theta = 1$
8. $c^2 = a^2 + b^2 - 2ab \cos C$
9. $[r(\cos \theta + i \sin \theta)]^n = r^n (\cos n\theta + i \sin n\theta)$
10. $S = a_1 / (1 - r)$